



DECEMBER 2024

MECHANICAL ENGINEERING MATERIALS*Time Allowed: 3 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from the rest.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (Any ten): 2x10=20

- i) An alloy of copper and zinc is known as- (a) Brass (b) Monel metal (c) Bronze (d) Hindalium.
- ✓/ii) German silver contains- (a) 70% Cu, 30% Zn (b) 60% Cu, 40% Zn (c) 40% Zn and 1% tin (d) 60.45% Cu, 35.2% Zn, 5.35% Ni.
- ✓/iii) When the austenitic steel (above 723°C) is cooled in air, then the resulting structure will be- (a) Fine pearlite (b) Martensite (c) Coarse pearlite (d) None of these.
- iv) Which alloying element is added to cast iron mainly with a view to mitigate the ill effects of sulphur?
(a) silicon (b) phosphorus (c) manganese (d) molybdenum.
- ✓/v) Which of the following is basically used in forging purpose? (a) Medium carbon steel (b) High speed steel (c) High carbon steel (d) Mild steel.
- ✓/vi) Which of the following alloy is mainly used for aircraft application? (a) Bronze (b) Babbitt metal (c) Duralumin (d) None of these.
- vii) Fibre reinforced plastic is- (a) alloy (b) ceramic (c) composite (d) metal.
- viii) The property which is desirable in a material subjected to shock loading – (a) Ductility (b) Malleability (c) Hardness (d) Toughness.
- ix) Cast iron is a product of- (a) Cupola (b) Bessemer converter (c) Open hearth furnace (d) Blast furnace.
- x) Red hot hardness of an alloy steel can be improved by adding - (a) Tungsten (b) Cobalt (c) Manganese (d) Both (a) & (b).
- xi) Which of the following is the hardest? (a) Austenite (b) Pearlite (c) Bainite (d) Martensite.

- xii) Which of the following is/are an example of Thermo-setting plastic? (a) Melamine (b) Epoxies (c) Bakelite (d) All of these.
- xiii) Low cost camera lenses are made by - (a) Acrylics (b) Nylon (c) ABS (d) None of these.
- xiv) The smallest part of a crystal lattice is known as - (a) Lattice structure (b) Lattice point (c) Bravais lattice (d) Unit cell.
- xv) The lower critical temperature (in °C) in iron-carbon phase diagram is - (a) 1539 (b) 1143 (c) 910 (d) 723.

GROUP-B

- 2) (a) Why tempering is required in Heat Treatment operation?
(b) Explain Martempering process with suitable TTT diagram. (2+6)
- 3) (a) Classify Plain carbon Steels along with their composition and applications.
(b) Write down the composition and use of wrought iron. (6+2)
- 4) (a) Write down the advantages, limitations of Ultrasonic Crack Detection and its area of application. <https://www.wbscteonline.com>
(b) State the properties and application of ABS and polyesters. (4+4)
- 5) (a) What is the meaning of 'XT 72 W 18 Cr 4 V1'?
(b) Identify different alloying elements along with their effect on steel. (2+6)
- 6) (a) State the common properties of plastics.
(b) Explain briefly the working principle of Plastic Injection moulding process with neat sketch. (2+6)
- 7) (a) Describe Allotropy of Iron.
(b) Draw Iron-Carbon equilibrium diagram indicating different phases, their carbon percentage and temperature of transformation. (2+6)
- 8) (a) Write down the property, composition and use of Y-alloy and Gun metal.
(b) Describe Flame Hardening process. (5+3)
- 9) (a) Write down the Coordination Number and Atomic Packing Factor for SC, BCC and FCC.
(b) Specify two important methods to prevent the corrosion. (6+2)